

JKBOSE 2026

CLASS 11TH

Bhai chare
ON
TOP

ORGANIC CHEMISTRY: SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-14 CHEMISTRY

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Topics To Covered

1

Methods of Purification of Organic Compounds

Part 2

2

Qualitative analysis

Methods of Purification

a) SUBLIMATION

b) CRYSTALLISATION.

c) DISTILLATION

d) Differential Extraction

e) Chromatography.

Fractional (5-10 k).

Reduced pressure (decompose before BP)

Steam distillation (BP very low)

Methods of Purification of Organic Compounds

(iv) STEAM DISTILLATION: (BP is very low) (Those liquids which are water insoluble)

- This method is used for separation and purification of organic compounds (solids or liquids) which
 - (a) Are steam volatile
 - (b) Are insoluble in water.
 - (c) Possess high vapour pressure (10–15 mm of Hg at 373 K)
 - (d) Contains non volatile impurities.
- Compounds which can be purified by steam distillation are aniline, nitrobenzene, bromobenzene, *o*-nitrophenol, *o*-hydroxy benzaldehyde (salicylaldehyde), *o*-hydroxy acetophenone, turpentine oil and essential oils.

$$BP \propto \frac{1}{VP}$$

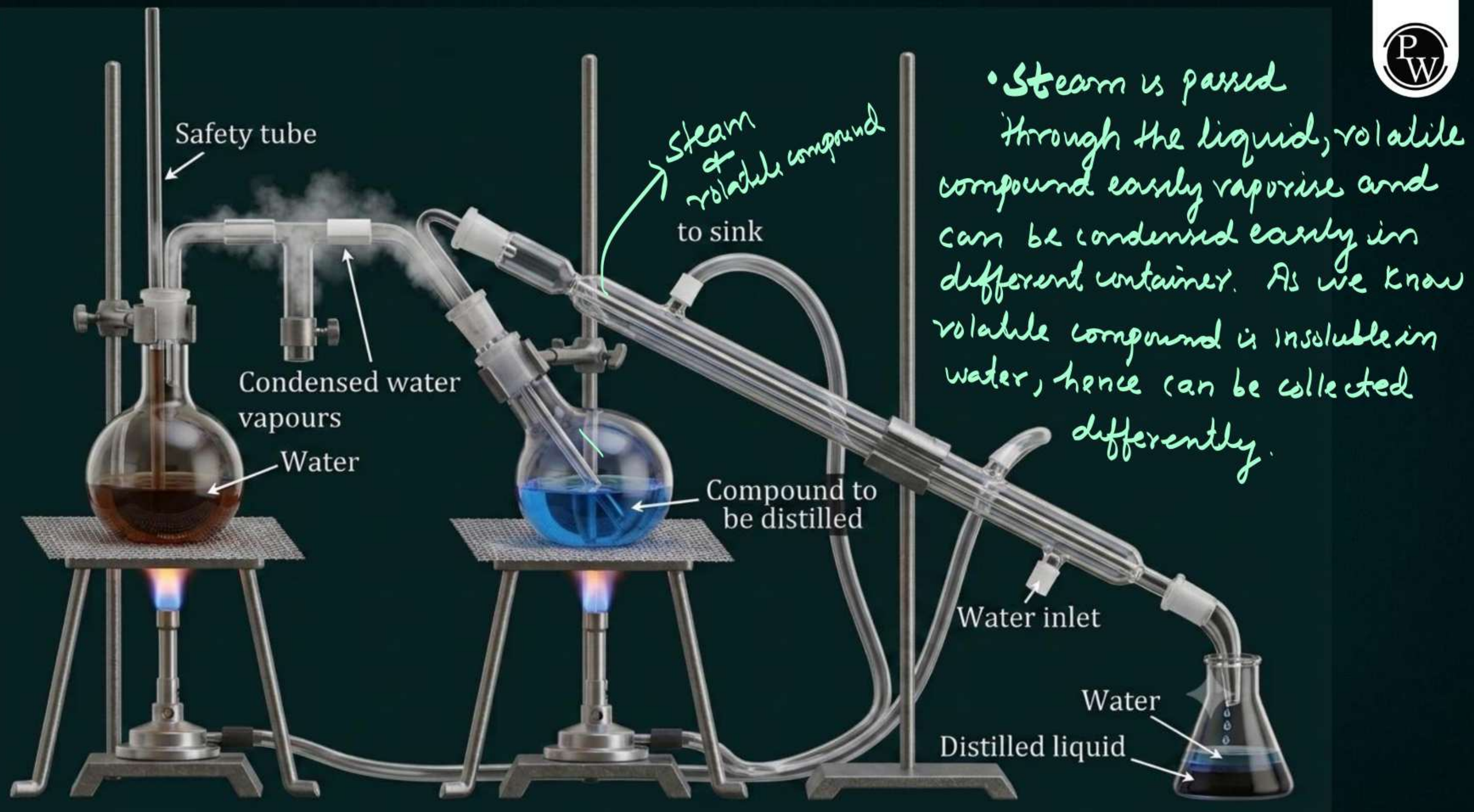


Figure: Steam distillation. Steam volatile component volatilizes, the vapours con.

Methods of Purification of Organic Compounds

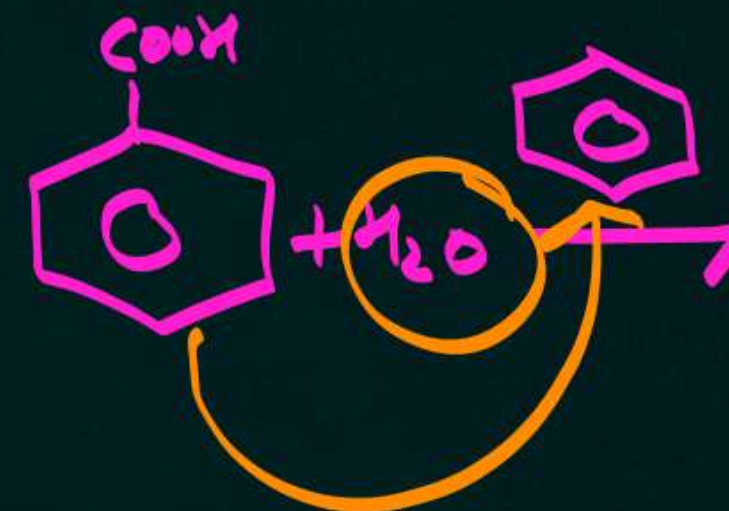
4. DIFFERENTIAL EXTRACTION: (Solubility)

Aqueous Soln of Compound
(H_2O + Compound) $\xrightarrow{\text{Solvent}}$

- The process of isolating an organic compound from its aqueous solution by shaking with a suitable solvent is called differential extraction. It is also called solvent extraction.
- ✗ When an organic compound is present in an aqueous medium, then it is separated by shaking it with an organic solvent in which it is more soluble than in water.
- Solvent should be immiscible with water and organic compound to be separated should be highly soluble in it.
- Organic solvent and aqueous solution are immiscible with each other, so they can form two distinct layers which can be separated by separatory funnel.
- Organic solvent is distilled or evaporated to get organic compound.

Methods of Purification of Organic Compounds

- Benzoic acid can be extracted from its aqueous solution using benzene as solvent.
- Ether is a better solvent in differential extraction due to:
 - (a) Its less polarity
 - (b) Least reactivity
 - (c) Higher solubility of organic compounds



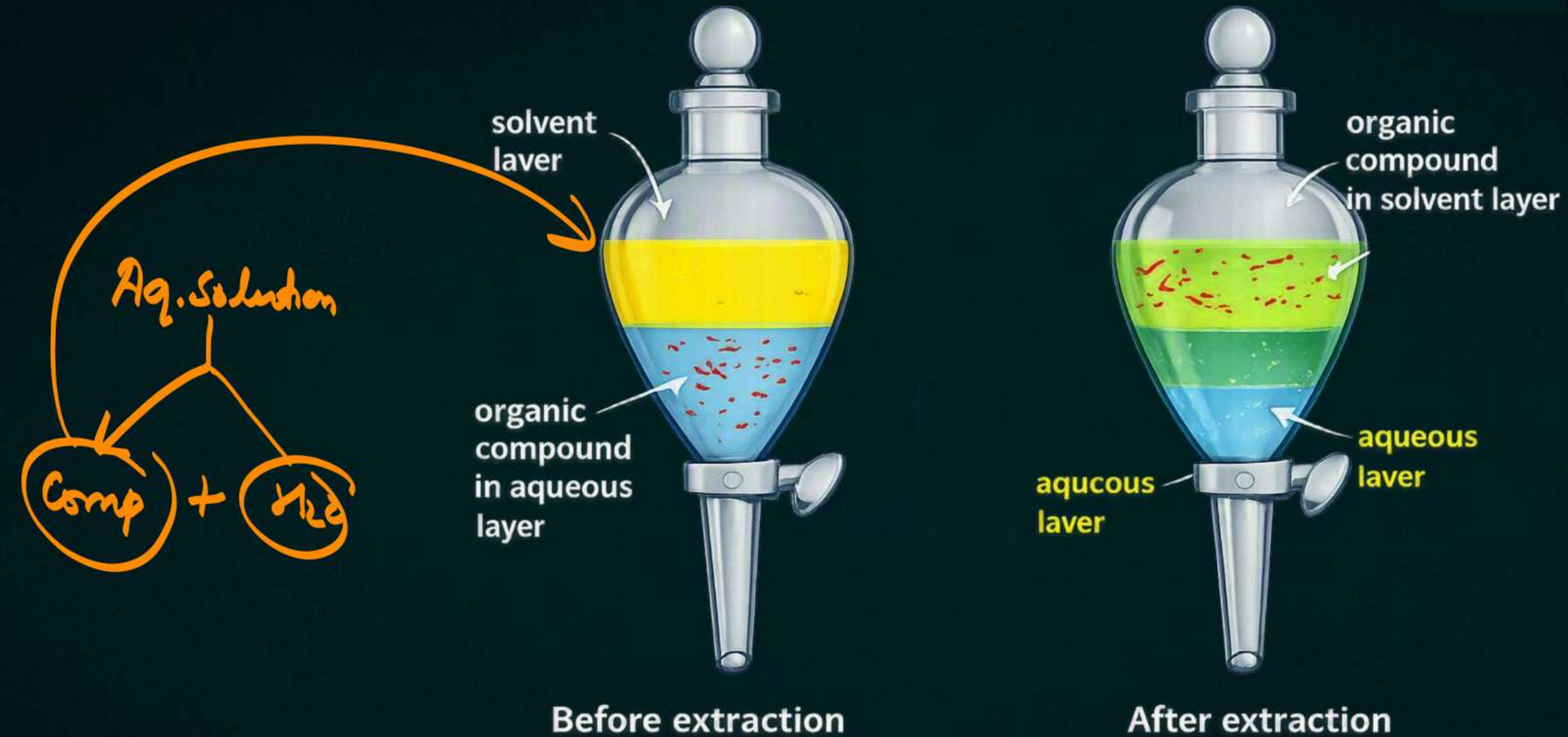


Figure: Differential extraction. Extraction of compound takes place based on difference in solubility

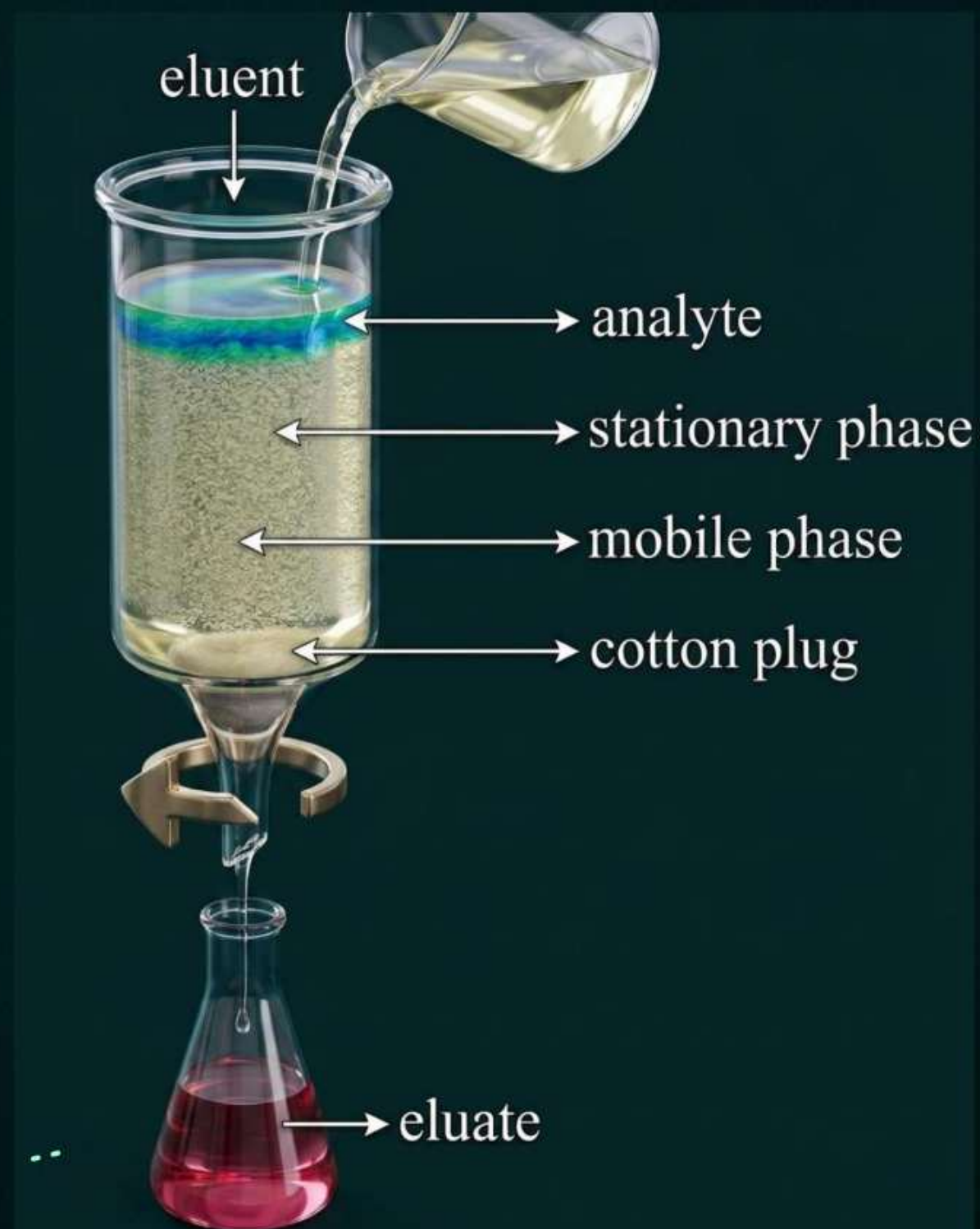


- 5) ➤ **Chromatography:** This method is widely used for separation, purification, identification and characterisation of the components of a mixture, whether coloured or colourless. The technique of separating the components of a mixture in which separation is achieved by the differential movement of individual components through a stationary phase under the influence of a mobile phase.
- **Types of chromatography:** Depending upon the nature of the stationary and the mobile phases, the different types of chromatographic techniques commonly used are in a given table.

Methods of Purification of Organic Compounds



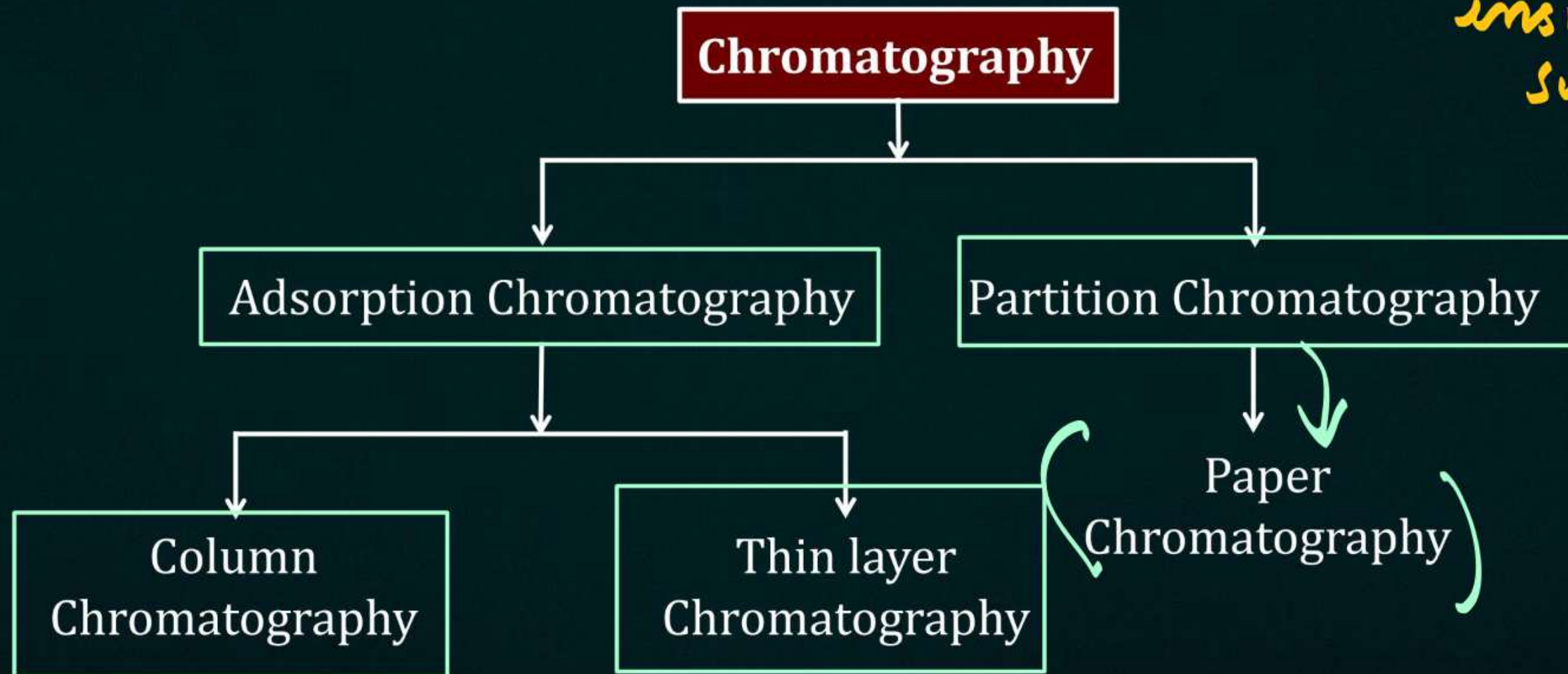
	Term	Definition
(a)	Mobile phase or carrier	Solvent moving through the column
(b)	Stationary phase or adsorbent	Substance that stays fixed inside the column
(c)	Eluent	Fluid entering the column
		Fluid exiting the column (that is collected in flask)
(d)	Elution	The process of washing out a compound through a column using a suitable solvent
(e)	Analyte	Mixture whose individual components have to be separated and analysed



Methods of Purification of Organic Compounds

Classification of Chromatography

Adsorption $\rightarrow$ *conc. of compd $\uparrow$ on the surface.*
 Absorption $\rightarrow$ *conc. of compound inside the surface $\uparrow$.*



- **Adsorption chromatography:** A chromatography in which different compounds are adsorbed on adsorbent at different degrees is called adsorption chromatography. Some well known adsorbents are silica gel, alumina, etc. Mobile phase is allowed to move on stationary phase. Two types of adsorption chromatography are **column chromatography** and **thin layer chromatography**.
- **Column chromatography:** It involves separation of mixture over a column of adsorbent packed in a glass tube.
 - The column is fitted with a stop cock at its lower end. The mixture adsorbed on adsorbent is placed on the top of the adsorbent column packed in a glass tube.
 - An appropriate eluent which is liquid or a mixture of liquids is allowed to flow down the column slowly.
 - The most readily adsorbed substance is retained near the top and the other comes down to various distances in the column.

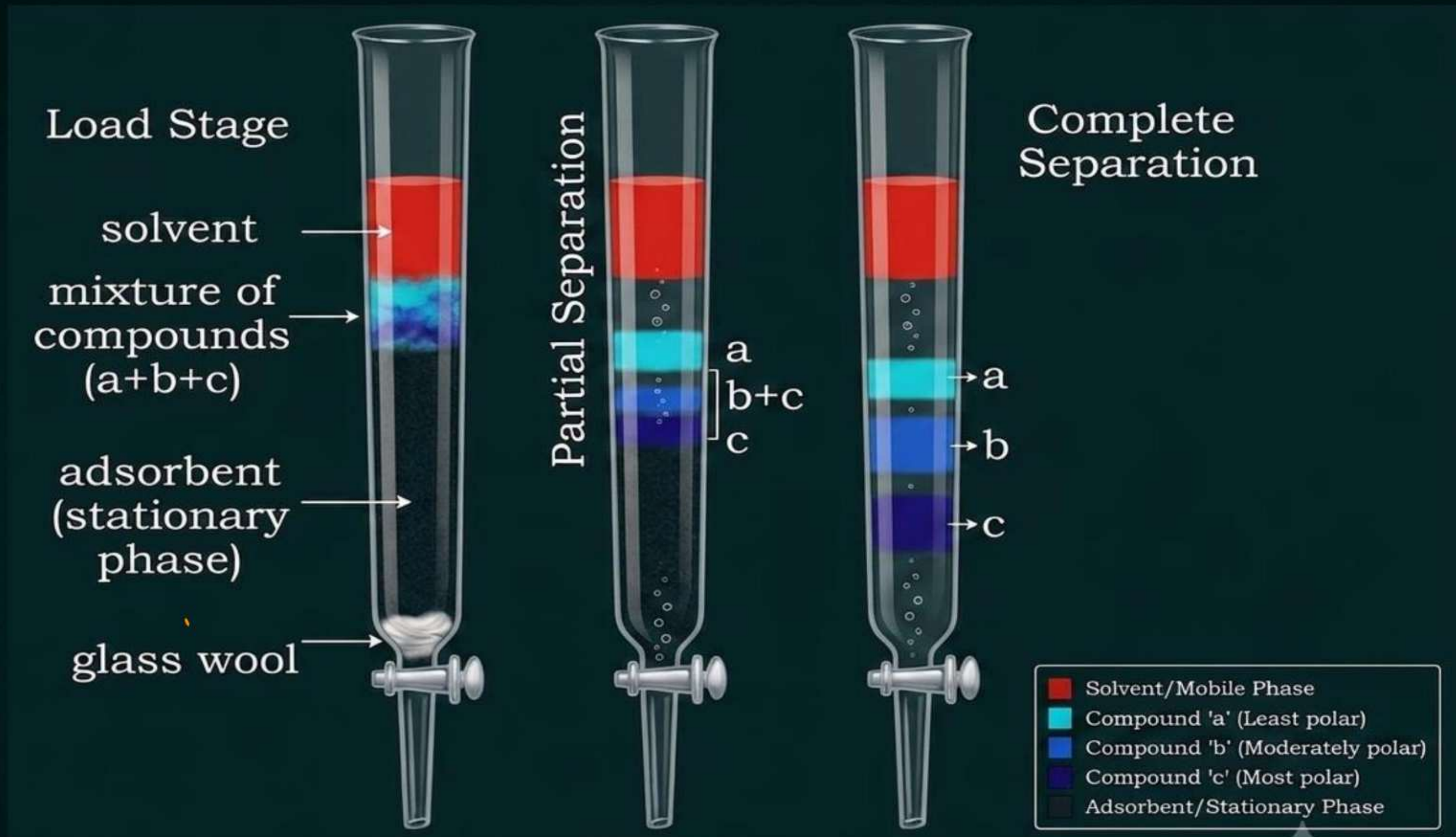


Figure: Column chromatography. Different stages of separation of components of a mixture.

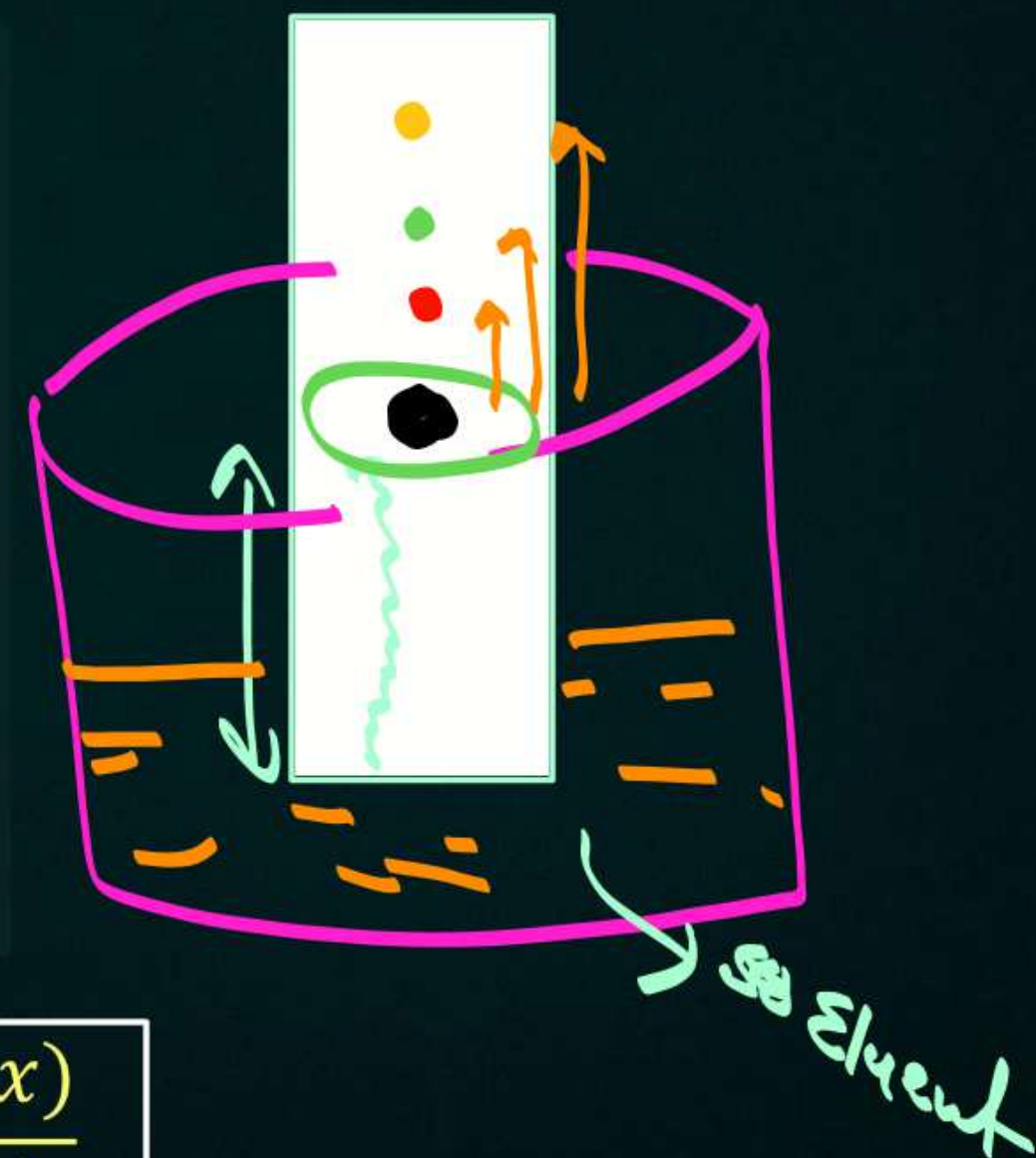
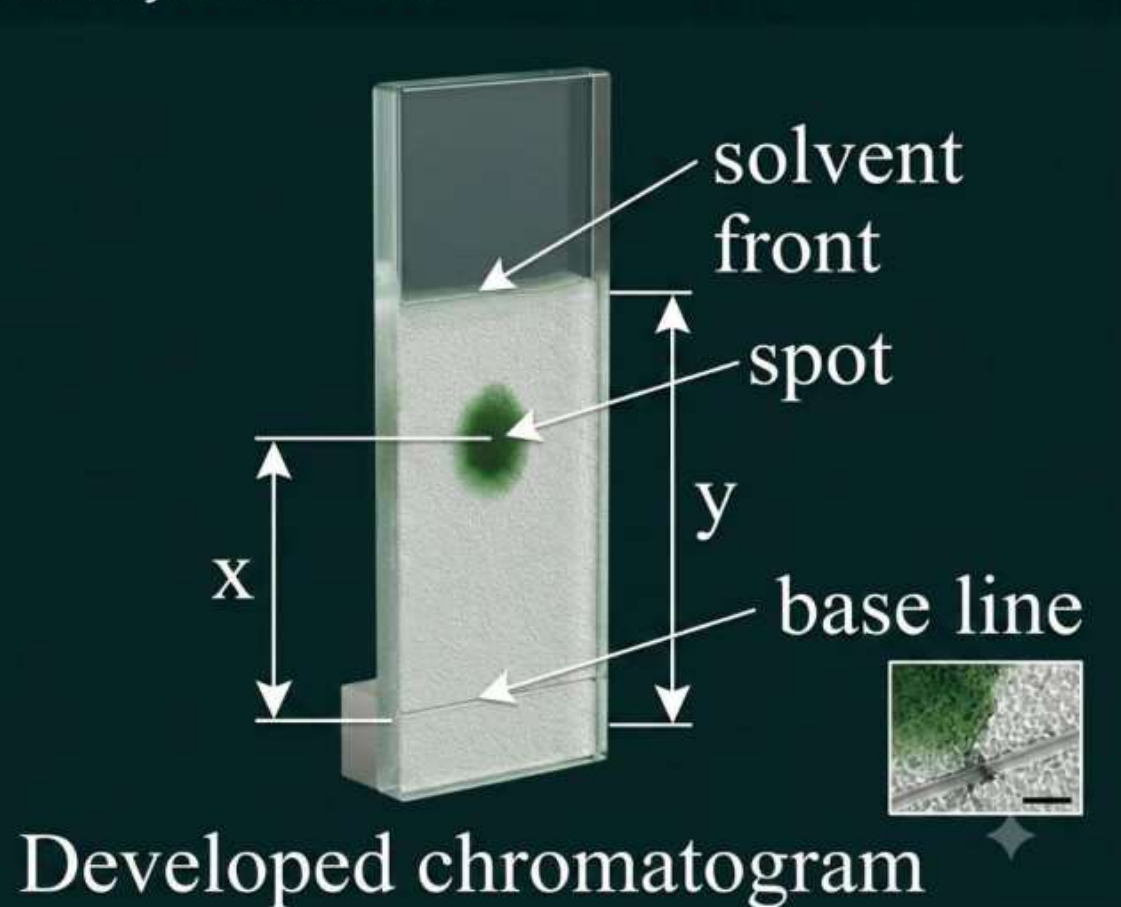
Methods of Purification of Organic Compounds

(ii) Thin Layer Chromatography

- Thin layer chromatography (TLC) is another type of adsorption chromatography, which involves separation of substances of a mixture over a thin layer of an adsorbent coated on glass plate.
- A thin layer (about 0.2mm thick) of an adsorbent (silica gel or alumina) is spread over a glass plate of suitable size.
- The plate is known as thin layer chromatography plate or chromaplate.
- The solution of the mixture to be separated is applied as a small spot about 2 cm above one end of the TLC plate.
- The glass plate is then placed in a closed jar containing the eluant.
- As the eluant rises up the plate, the components of the mixture move up along with the eluant to different distances depending on their degree of adsorption and separation takes place.

Methods of Purification of Organic Compounds

- The relative adsorption of each component of the mixture is expressed in terms of its retardation factor i.e. R_f value...



$$R_f = \frac{\text{Distance moved by the substance from base line (x)}}{\text{Distance moved by the solvent from base line (y)}}$$

Methods of Purification of Organic Compounds

- The spots of coloured compounds are visible on TLC plate due to their original colour.
- The spots of colourless compounds, can be detected by putting the plate under ultraviolet light.
- Another detection technique is to place the plate in a covered jar containing a few crystals of iodine. Spots of compounds, which adsorb iodine, will show up as brown spots.
- Sometimes an appropriate reagent may also be sprayed on the plate for example: Amino acids may be detected by spraying the plate with ninhydrin solution.

Methods of Purification of Organic Compounds

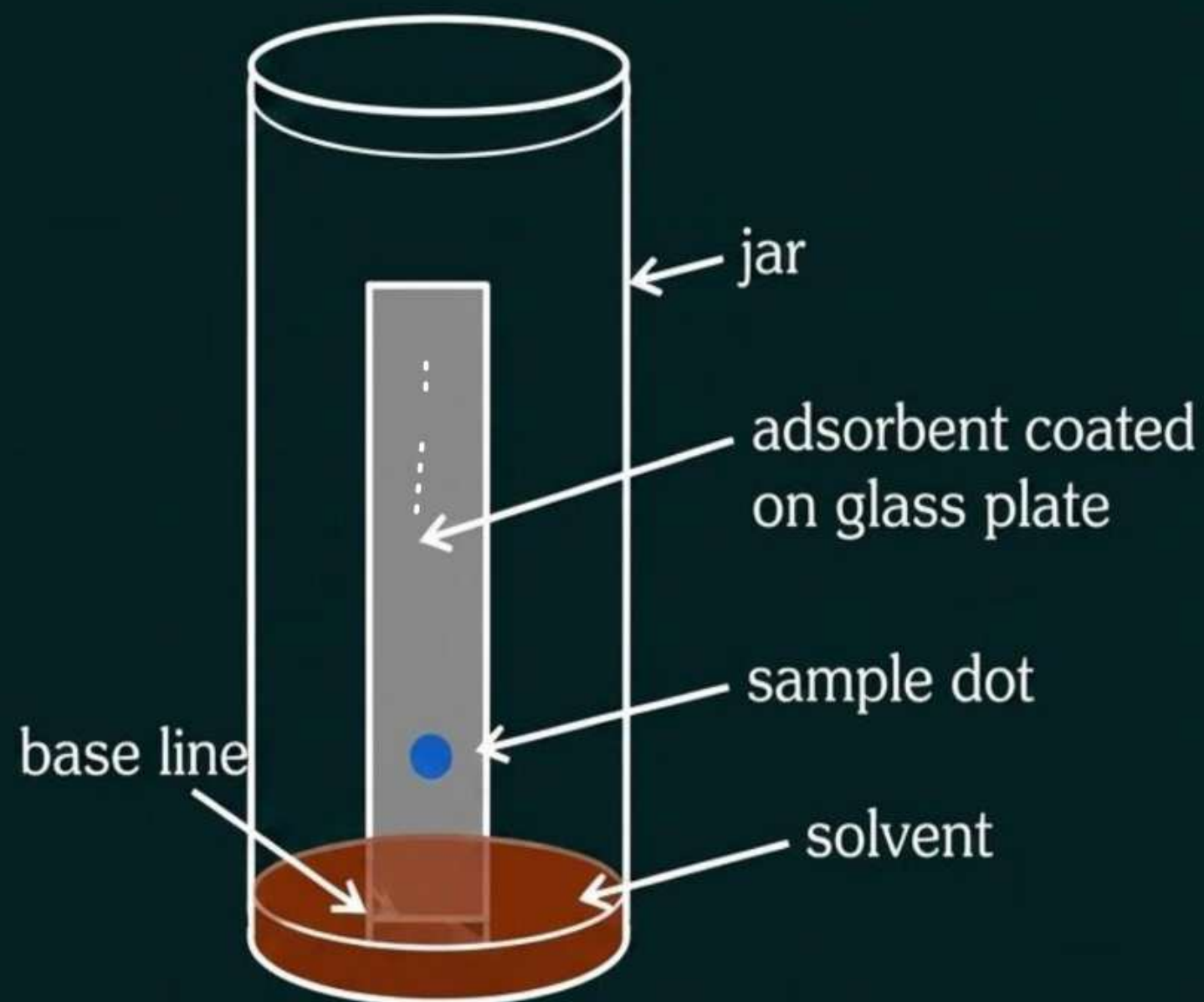


Figure: (a) Thin layer chromatography. Chromatogram being developed.

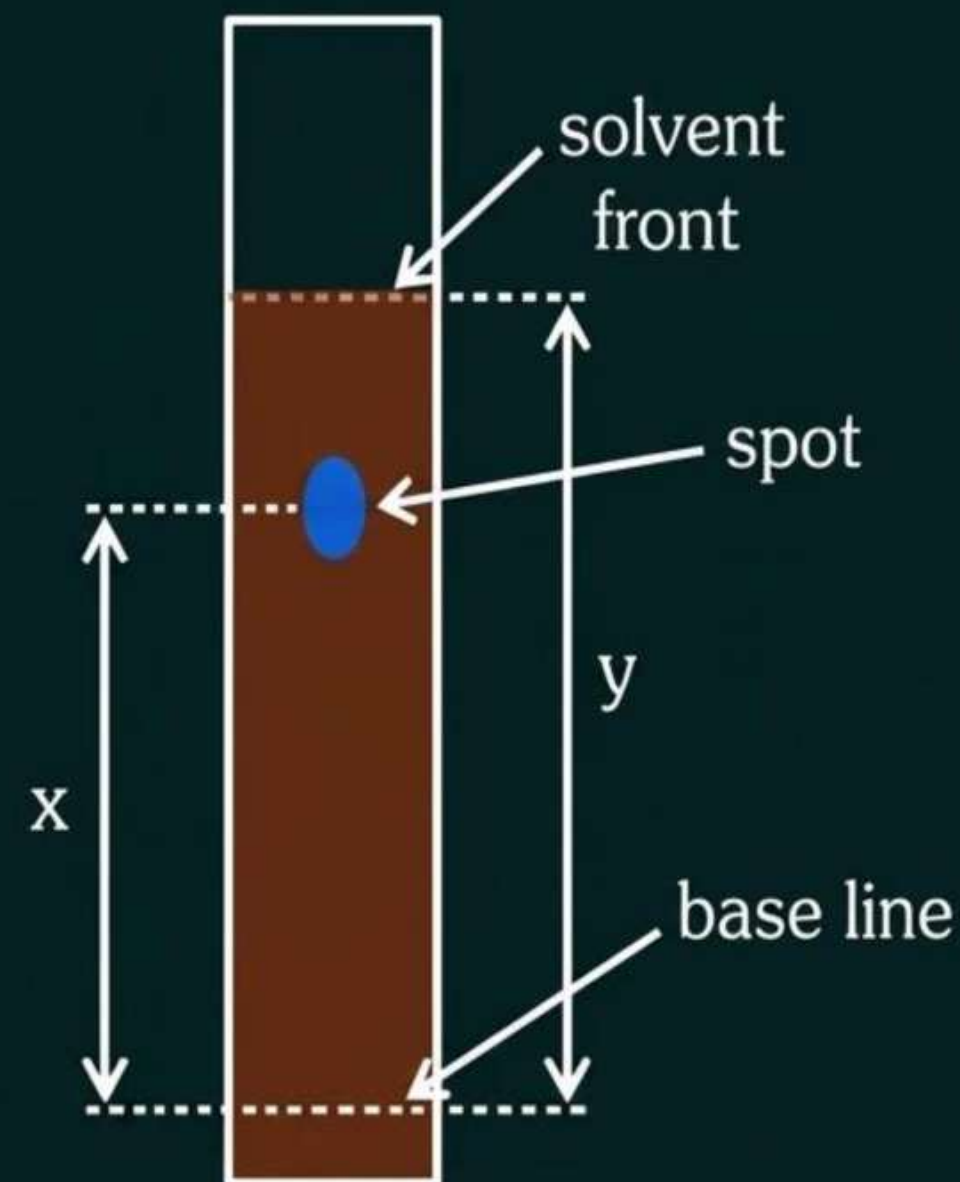


Figure: (b) Developed chromatogram.

Methods of Purification of Organic Compounds

(b) Partition Chromatography

- Partition chromatography is based on continuous differential partitioning of components of a mixture between stationary and mobile phases...
- Paper chromatography is a type of partition chromatography.
- In paper chromatography, a special quality paper known as chromatography paper (made of cellulose and water is absorbed in it) is used.
- Chromatography paper contains water trapped in it, which acts as the stationary phase.
- A strip of chromatography paper spotted at the base with the solution of the mixture is suspended in a suitable solvent or a mixture of solvents.
- This solvent acts as the mobile phase.
- The solvent rises up the paper by capillary action and flows over the spot.

Methods of Purification of Organic Compounds

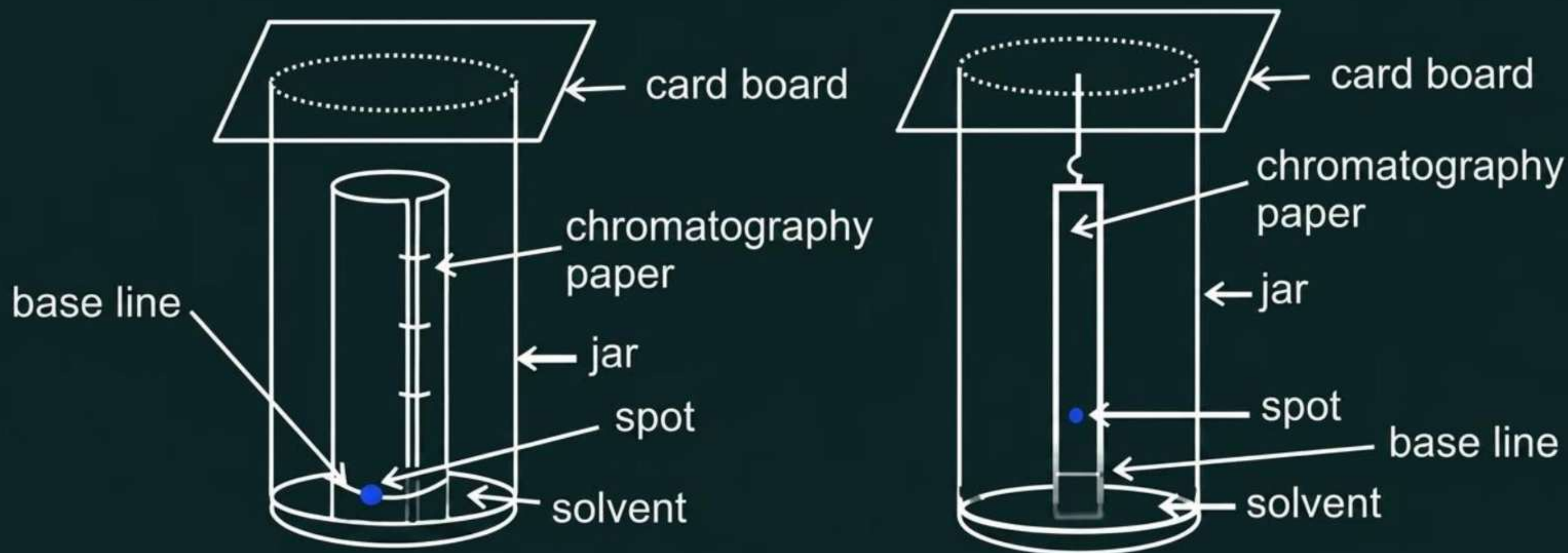


Figure: Paper chromatography. Chromatography paper is two different shapes.

Methods of Purification of Organic Compounds

- The paper strip so developed is known as a **chromatogram**.
- The spots of the separated coloured compounds are visible at different heights from the position of initial spot on the chromatogram.
- The spots of the separated colourless compounds may be observed either under ultraviolet light or by the use of an appropriate spray reagent as discussed under thin layer chromatography.

Same as TLC
 → TLC used glass
 → Paper used paper.

Methods of Purification

Qualitative Analysis

Qualitative And Quantitative Analysis of Organic Compound

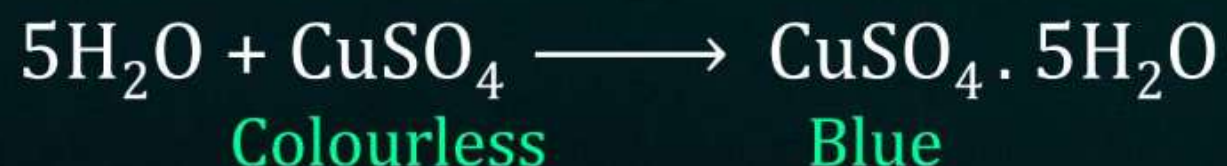


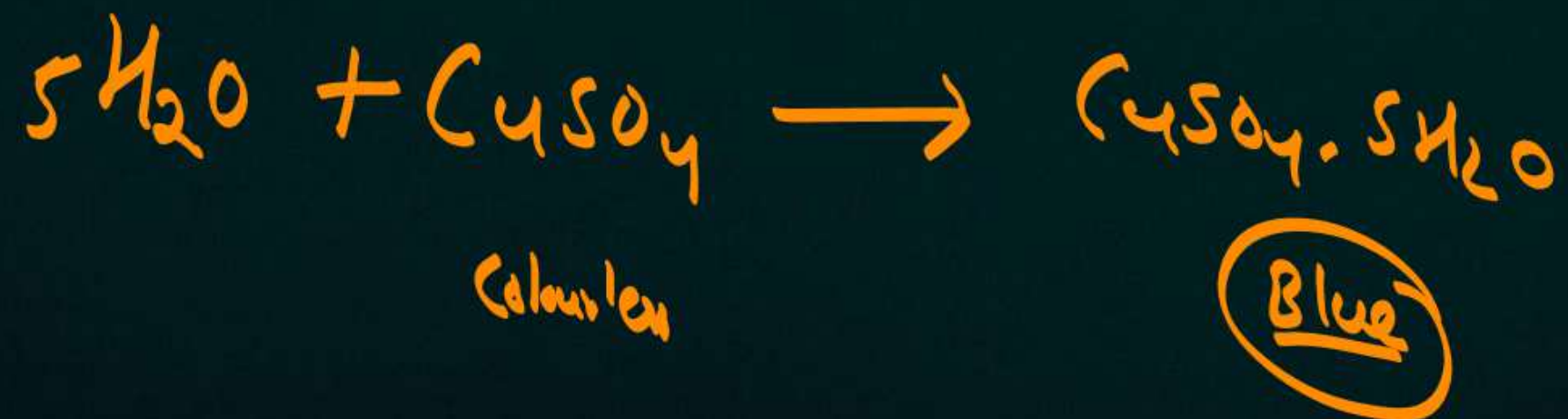
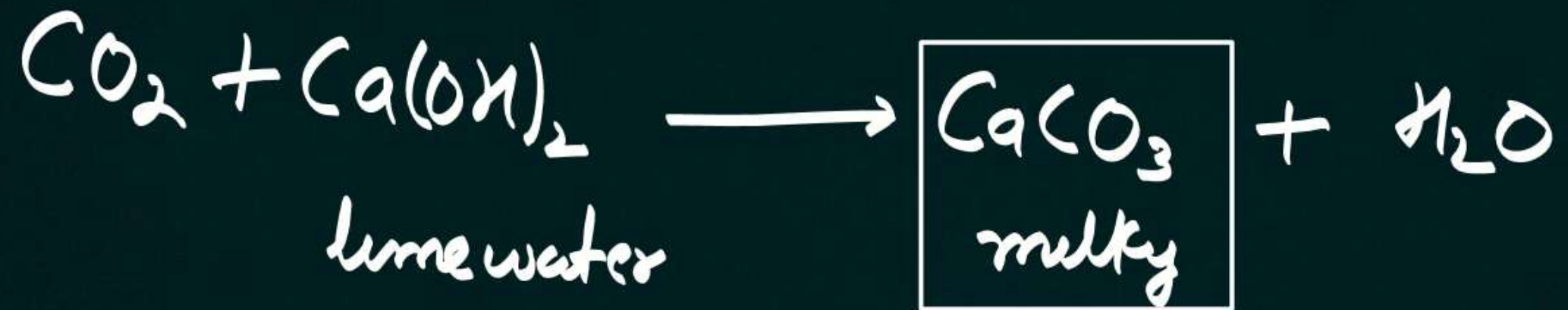
Reactant + Reagent $\longrightarrow$ Product

Qualitative analysis: Detection of element present in compound:

(i) Detection of C and H: Compound heated with cupric oxide and produced vapour is passed through lime water or anhydrous CuSO_4 . If lime water turn milky, it shows presence of CO_2 or carbon.

If anhydrous CuSO_4 turn blue than it shows presence of H_2O or hydrogen.,





QUESTION



Chromatography is used for the purification of:

- 1 Solids
- 2 Gases
- 3 Liquids
- 4 All of these

QUESTION



Chromatography technique is used for the separation of:

- 1 Small samples of mixtures
- 2 Plant pigments
- 3 Dye stuffs
- 4 All of the above

QUESTION



The compound that does not give a blue colour in Lassaigne's test is:

- 1 Aniline
- 2 Glycine
- 3 Hydrazine
- 4 Urea

QUESTION



In Lassaigne test thiourea is converted into:

- 1 NaCNS
- 2 Na_2S
- 3 NaCN
- 4 Na_2SO_4

QUESTION



Lassaigne's test is used in qualitative analysis to detect

- 1 Nitrogen
- 2 Sulphur
- 3 Chlorine
- 4 All of these

QUESTION

The sodium extract of an organic compound on acidification with acetic acid and addition of lead acetate solution gives a black precipitate. The organic compound contains.

- 1 Nitrogen
- 2 Halogen
- 3 Sulphur
- 4 Phosphorus

QUESTION

The prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of:

- 1 $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- 2 $\text{Na}_3 [\text{Fe}(\text{CN})_6]$
- 3 $\text{Fe}(\text{CN})_3$
- 4 $\text{Na}_4(\text{Fe}(\text{CN})_5\text{NOS})$

Q. What do you mean by

3 Marks?

a) Chromatography.

b) Differential extraction

Thank
You